

Classes, Objects and Inheritance

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5 / 5 submitted

Q1. Program based on Classes and Objects

Score: 3.33 TC: 2/3 passed

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        Student a=new Student();
        a.rollNo=sc.nextInt();
        sc.nextLine();
        a.name=sc.nextLine();
        a.marks=sc.nextDouble();
        a.displayDetails();
    }
}
class Student{
    int rollNo;
    String name;
    double marks;
    public void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
}
```

INPUT

```
1
Dhilip
77
```

OUTPUT

```
Roll No: 1
Name: Dhilip
Marks: 77.0
```

Q2. Initialize Object Data Using Setter Methods

Score: 5.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s=new Student();
        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();
        s.setRollNo(rollNo);
        s.setName(name);
        s.setMarks(marks);
        s.displayDetails();
    }
}
class Student{
    private int rollNo;
    private String name;
    private double marks;
    public void setRollNo(int rollNo){
        this.rollNo=rollNo;
    }
    public void setName(String name){
        this.name=name;
    }
    public void setMarks(double marks){
        this.marks=marks;
    }
    public void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
}
```

INPUT

```
1
Dhilip
98.0
```

OUTPUT

```
Roll No: 1
Name: Dhilip
Marks: 98.0
```

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student a=new Student();
        a.displayDetails();
        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();
        Student b=new Student(rollNo,name,marks);
        b.displayDetails();
    }
}
class Student{
    int rollNo;
    String name;
    double marks;
    Student(){
        rollNo=101;
        name="Rahul";
        marks=85.5;
    }
    Student(int rollNo,String name,Double marks){
        this.rollNo=rollNo;
        this.name=name;
        this.marks=marks;
    }
    public void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
}
```

INPUT

```
1
Dhilip
88.0
```

OUTPUT

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 1
Name: Dhilip
Marks: 88.0
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Salary s=new Salary();
        s.empId=sc.nextInt();
        sc.nextLine();
        s.empName=sc.nextLine();
        s.basicSalary=sc.nextDouble();
        s.calculateSalary();
        s.displayDetails();
    }
}
class Employee{
    public int empId;
    public String empName;
    public double basicSalary;
}
class Salary extends Employee{
    public double hra;
    public double da;
    public double netSalary;
    public void calculateSalary(){
        hra=basicSalary*20/100;
        da=basicSalary*10/100;
        netSalary=(basicSalary+hra+da);
    }
    public void displayDetails(){
        System.out.println("Employee ID: "+empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic Salary: "+basicSalary);
        System.out.println("HRA: "+hra);
        System.out.println("DA: "+da);
        System.out.println("Net Salary: "+netSalary);
    }
}
```

INPUT

```
1
Dhilip
10000.0
```

OUTPUT

```
Employee ID: 1
Employee Name: Dhilip
Basic Salary: 10000.0
HRA: 2000.0
DA: 1000.0
Net Salary: 13000.0
```

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        GrossSalary g=new GrossSalary();
        g.empId=sc.nextInt();
        sc.nextLine();
        g.empName=sc.nextLine();
        g.basicSalary=sc.nextDouble();
        g.calculateSalary();
        g.displayDetails();
    }
}
class Employee{
    public int empId;
    public String empName;
}
class Salary extends Employee{
    public double basicSalary;
}
class GrossSalary extends Salary{
    public double hra;
    public double da;
    public double GrossSalary;
    public void calculateSalary(){
        hra=basicSalary*20/100;
        da=basicSalary*10/100;
        GrossSalary=basicSalary+hra+da;
    }
    public void displayDetails(){
        System.out.println("Employee ID: "+empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic Salary: "+basicSalary);
        System.out.println("HRA: "+hra);
        System.out.println("DA: "+da);
        System.out.println("Gross Salary: "+GrossSalary);
    }
}
```

INPUT

```
1
Dhilip
10000
```

OUTPUT

```
Employee ID: 1
Employee Name: Dhilip
Basic Salary: 10000.0
HRA: 2000.0
DA: 1000.0
Gross Salary: 13000.0
```